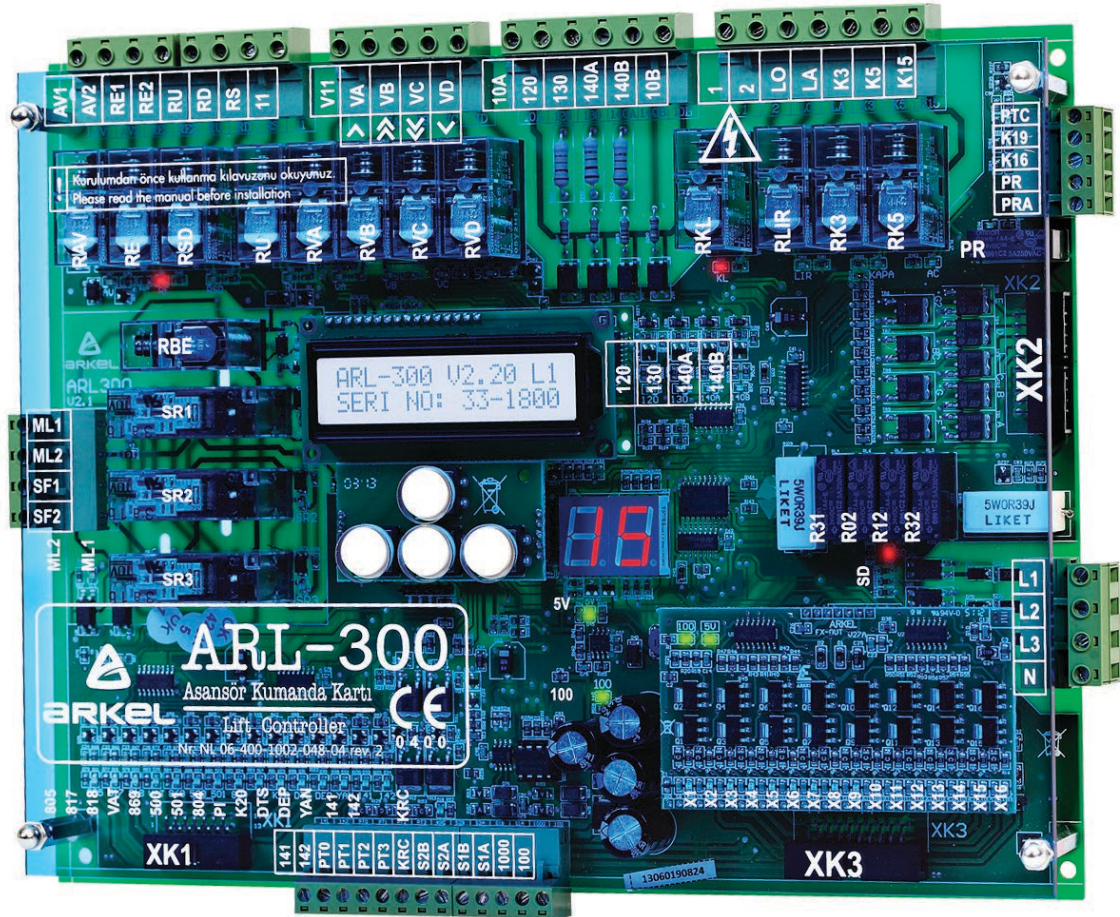




# ARL-300

## LIFT CONTROLLER

### ERROR CODES POSSIBLE REASONS AND SOLUTIONS

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## 1. ERROR CODES POSSIBLE REASONS and SOLUTIONS

### H1 FAIL TO LOCK DOOR. 140 SIGNAL IS MISSED:

After door close command is given by controller, if 140 signal is still not activated within the value at “door lock wait” parameter, this error is raised.

- If the parameter is too low, it should be increased.
- Safety circuit should be checked.

### H2 FLOOR TO FLOOR MAX TRAVEL TIME ELAPSED:

If the car could not reach a new floor within the value at “Max. Floor to floor” parameter, this error is raised.

- The value in the parameter may be too low, you can increase it.
- If the parameter is enough, you should check that ML1 and ML2 signals are activated in the door zone during the movement.
- If inverter is kept in the menu, and give a call, after a while it will give this error, since controller has activated the relay.

### H3 DOOR IS NOT CLOSED FOR A LONG TIME:

If the door is not closed (140 signal) within the value at “door open error” parameter, this error is given and system goes to out of service.

- You should increase the parameter not to receive the error constantly.
- Although the door is closed, if still you receive this error, you can check door contacts and 140 signal.

### H4 MAX SLOW SPEED TRAVEL TIME ELAPSED:

If the car goes with slow speed during the value at “Max. Low speed time” parameter, this error is raised.

- You should increase it up to an available value with trial and error.
- If the parameter is enough, you should check that ML1 and ML2 signals are activated in the doorzone during slow speed movement.

### H5 BOTH LIMIT SWITCHES UP&DOWN IS OPEN:

Normally, at most, only one of the limit-switches can be open (zero) and other one must be close (one) at any time. If both signals are open same time this error is raised. The limit switches and their magnets should be checked.

### H6 TOP LIMIT SWITCH CUTS OFF WHILE RUNNIG DOWN:

You should check the round magnets red and black sides. 818 magnetic switch is used in the top floor and can be cut only when the car arrives at top floor. Otherwise, it gives error.

### H7 BOTTOM LIMIT SWITCH CUTS OFF WHILE RUNNIG UP:

You should check the round magnets red and black sides. 817 magnetic switch is used in the bottom floor and can be cut only when the car arrives at bottom floor. Otherwise, it gives error.

### H8 MOTOR PTC CIRCUIT IS OPEN. MOTOR OVERHEAT:

The terminal PTC on ARL300 monitors the heat. There should be 24Vdc here. If not, this error is triggered. Check PTC circuit in order to understand which contact is open.

### H9 SYSTEM ERROR

Please get contact to Arkel technical support.

### H10 CONTACTOR CHECK BACK ERROR

You should check KRC signal on ARL300. While the car is stopping, this input must be ON. While moving, it should be OFF. Otherwise, you should check the NC contacts of the contactors. There is another option, you can perform this function with VVVF inverter. “Contactor Check” parameter can be chosen as “ARL300” or “ADrive”. If ADrive is chosen, EN signal is used for the monitoring function. (This is not a permanent error)

### H11 ML1 & ML2 SIGNALS SHORT CIRCUIT TO EACH OTHER:

ML1 and ML2 magnetic switches should be ON or OFF with a delay. It is not allowed to be active or passive together at the same time. If this case occurs, this error will be received. You should check the distance between the magnetic switches. It should be 5 cm. The distance from magnetic switch to door-zone magnet should be 1-2 cm. If REVBA board is used on the car roof, the common of magnetic switches must be connected to ML0. This input is supplied over battery. After energy is OFF, here still will have voltage. Then, it will be prevented to receive error as soon as the energy is back again. ML1 and ML2 positions must be checked. The ML2 switches is superposed on ML1 switch. If the car moves downward firstly ML1 signal must be energized before ML2 when car reaches floor level. If it is opposite the error is raised. If the sequency of ML1 and ML2 is correct and

connections are correct, the 1000 supply of inspection box should be done as double connections. A resistance(1kohm/2W) should be put between 1000 and ML1 also between 1000 and ML2 to get magnetic noise on board.

**H12 DOOR BRIDGING RELAYS SR2 & SR3 DONT PICK UP**

You should check ML1-ML2.

**H13 DOOR BRIDGING RELAYS SR2 or SR3 DONT DROP**

You should check ML1-ML2.

**H14 DOOR BRIDGING RELAY SR1 CHECK BACK ERROR**

When the door bridging is terminated, it is expected that 140 signal is cut. (turns to OFF position)

You should be sure that there is no bridge in the safety circuit and the connections must be connected according to Arkel safety circuit diagrams.

**H15 RST MAINS POWER SUPPLY FAILURE:**

ARL300 cannot detect the 3-phase from L1-L2-L3 terminals. You should measure them on AC mode. If 3-phase is coming, the components on the controller for phase detection may have a problem.

**H16 24V (100) SUPPLY IS UNDER LIMIT:**

100-1000 voltage is low. It should be almost 22-24Vdc under normal conditions. You can check the output of isolation transformer and bridge rectifier. If everything is as expected, there may be a problem with the capacitor on ARL300.

**H17 PTC SIGNAL OPEN. MOTOR OVER HEATED**

The terminal PTC on ARL300 monitors the heat. There should be 24Vdc here. If not, this error is triggered. Check PTC circuit in order to understand which contact is open.

**H18 FAULT SIGNAL FROM VVVF DRIVER:**

The fault signal from inverter comes to controller. If inverter gives error, control detects it, and gives this error. You should focus on the error on inverter. If inverter does not give any error, but still, you see H18 on controller screen, you should check PI terminal on KBK4. Normally, there should be 24Vdc (not to receive an error)

**H19 SAME GROUP ID NAME USED ON CONTROLLERS:**

ID numbers in the group must be different from each other. If one of them is Controller-A, other one should be Controller-B etc... It should be adjusted from the parameter menu correctly.

**H20 120 (STOP) SIGNAL IS MISSING. ELEVATOR BLOCKED:**

120 signal is cut because of a reason. Under normal running conditions 120 is not cut. (Even if doors are opened, 120 must be ON position)

- At the output of isolation transformer 220Vac must be measured.
- You should check the safety circuit in order to find which contact is open in the series circuit.

**H21 EXTERNAL SAFETY RELAY CHECK BACK ERROR:**

For Hydraulic Systems, PI input (on KBK4) is used to monitor external contactor checkback signal. In some systems which the valve block is controlled by an electronic board like Beringer, valve blocks are not supplied over the safety circuit. The safety circuit activates 2 different contactors. And NC contact of these contactors are connected in serial from 100 to PI input. Thus, contactor stick is monitored. If external contactors are used in hydraulic systems, PI is used for monitoring the up and down valves. This input works like KRC signal in traction lifts. While the car is stopping, there should be signal in this terminal. During movement, there should not be. Except these cases, check the external contactors and their contacts.

**H22 SERVICE TIME OVERFLOW:**

When this error is encountered, "service time" parameter should be changed. "End of service time" is adjusted from the parameter as "continue working" or "stop".

**H23 A3 HDU VALVE TEST ERROR:**

It seems that in those ARL-300 controllers' parameter "UCM Safety Valve" is set to "Double Valve chk". This option can be applied in case of using two electrically commanded hydraulic valves operating in series. Because the A3 safety valve and downward travel are energized at the same time, a self-monitoring is required. And this error means ARL-300 controller detects a failure during self-monitoring test.

**H24 UCM DETECTION ERROR:**

- The distance of door zone magnetic switches should be checked. The distance should be 5 cm between ML1 and ML2 magnetic switches.
- Door preopening and re-levelling should be disabled from the menu of ARL300.



**H25 CONTACTOR CHECKBACK ERROR:**

If "H10 contactor check back" error repeats 3 times (see H10 error explanation), it goes to a permanent error and H25 error code seems in the screen. This error blocks the elevator. When this error is encountered, the "fault list" should be cleared from the parameter. Then, the elevator will turn to normal run mode. The contactors and its NC (normally) contacts should be checked. While the car is stopping, it should have signal. While it is moving, it should not have.

**H26 MOTOR BRAKE MONITORING:**

Brake micro-switches in gearless machines can be monitored over 805 input directly on ARL300 board (if and only if FX-Seri board exists, and it is not a firefighter lift). Brake monitoring parameter must be enabled from menu. While the car is stopping, the input must be ON. (24Vdc) While the car is moving, this should be OFF. Otherwise, it gives this error. It is not a permanent error. If you do not do brake monitoring from controller, you should disable it from menu.

**H27 MOTOR BRAKE MONITORING (PERMANENT):**

When H26 occurs 2 times consecutively, this permanent error is raised.

- When this error is encountered, the "fault list" should be cleared from the parameter. Then, the elevator will turn to normal run mode.
- If you do monitoring from controller, the brake micro-switches and its connections should be checked. (While the car is stopping, it should have signal. While moving, it should not.)
- If you do monitoring over inverter, you should disable monitoring function from ARL300 controller.

**H28 DOOR BRIDGED ERROR:**

When the elevators stop at floor, when "door open" is activated, if safety circuit (130-140) is not interrupted, this error occurs. There is a counter inside. After this problem occurs 5-6 times, this permanent error is shown on the screen. You should clear the fault list, and you should check the safety circuit connections. You should observe that when the doors are opened, safety circuit signals must be cut.

**H30 ML1-ML2 SEQUENCE WRONG OR COULD NOT BE READ:**

If ML1 and ML2 state changes were in an unexpected sequence, this error is given.

- ML1 or ML2 is not connected or faulty
- A high voltage line might be passing next to ML1 or ML2
- ML1 or ML2 may be connected in the wrong order.
- Check if the placement of magnets related to ML1,ML2 signals are correct by running the car in inspection mode. (i.e: When moving slowly, observe ML1,ML2 signals. When moving upwards, on arrival of a door-zone, first ML2 then ML1 should activate. When departing a door-zone, first ML2 then ML1 should deactivate. And the opposite must happen when moving downwards)

**H31 THE NUMBER OF FLAGS LEARNED IS NOT CORRECT:**

The door zone magnet mounted and the parameter "Number of floors" should be arranged the same. If the parameter "Number of floors" is arranged to 5 but 4 or 6 door zone magnets are put, this error is given.

Check the required parameter and the door zone magnet you put.

**H32 817/818 FORCED LIMIT SWITCHED TURNED IN THE WRONG DIRECTION:**

If bottom limit switch (817) is interrupted while moving up except bottom floor or top limit switch (818) is interrupted while moving down except top floor, this error is given.

- Check bottom limit switch SKSR1 and SKSR2 mechanically or electrically.
- Check bottom limit switch input 817 and 818

**H33 WRONG ENCODER DIRECTION:**

If the travel distance is increasing during downward travel or if the travel distance is decreasing during upward travel, this error is given. Please check the encoder connection. You can change the A and B sequence.

**H34 NO ENCODER SIGNAL:**

Please check the encoder connection between ARL300 and ADrive.

- OA terminal of ADrive should be bridged to A- of ARL300.
- OB terminal of ADrive should be bridged to B- of ARL300.
- A and B terminals should be connected to +15 terminal of ADrive.

**H35 UCM SWITCH MONITORING:**

PI2 input of ARL300 board is used as over speed governor UCM switch monitoring input. You should connect UCM switch to PI2. Please, check connections.

If the over speed governor has no switch (not A3 standard), you should arrange PR2 Relay is “second motor control”.

**NOTE:** "H35" is permanent error and is not cleared even after the mains power is cut and back again. To clear the error: Enter the menu. Go to “erase fault” parameter.

### **H36 TOP LIMIT SWITCH ERROR:**

According to the European standard, in hydraulic elevators, when the car reaches the upper limit switch, it should be taken out of service. And it shall not resume normal operation until the elevator reaches the lower floor after the safety circuit returns.

- To avoid this error, the position of the upper limit switch, the deceleration and the stopping distance must be checked.
- Please also decrease the motor stop delay setting. First press the up button and then down button.